CLINICAL · APPROACH TO

Approach to a swollen salivary gland

A swollen salivary gland is either inflamed, obstructed, infiltrated or growing a tumour. The history usually separates these: swelling that comes with meals and goes again is obstruction; acute painful swelling with fever is infection; bilateral painless swelling is systemic; and a persistent discrete lump is a neoplasm until a needle says otherwise. Ultrasound with a fine-needle or core biopsy is the workhorse of investigation, and facial weakness with a parotid lump is malignancy until proven otherwise.

AT A GLANCE

- Meal-related swelling: stone or stricture.
- Bilateral diffuse: mumps, Sjögren, sialosis, HIV, sarcoid.
- Discrete lump: ultrasound and FNA.
- Children: parotid lumps are more often malignant than in adults.

1 CHAPTER ONE The pattern

Time course and side.

1.1 Sorting the swelling

SALIVARY SWELLING BY PATTERN

PATTERN	THINK OF	CLUES
Recurrent, swelling at meals	Stone or stricture	Submandibular more than parotid; pain with eating, settles in hours
Acute, painful, red, fever	Acute bacterial sialadenitis	Elderly, dehydrated, postoperative; pus from the duct
Acute, bilateral, with malaise in a child or young adult	Mumps or other viral parotitis	Contacts; unvaccinated; orchitis

PATTERN	THINK OF	CLUES
Recurrent parotid swelling in a child	Juvenile recurrent parotitis	Episodes settle by puberty; sialectasis on ultrasound
Bilateral, painless, persistent	Sialosis, Sjögren's syndrome, IgG4-related disease, sarcoidosis, HIV	Dry eyes and mouth; alcohol, diabetes; cysts
Discrete persistent lump	Neoplasm	Painless, slow (benign); pain, fixity, palsy, nodes (malignant)

RED FLAGS FOR MALIGNANCY

- **Facial nerve weakness** with a parotid lump.
- **Pain, rapid growth, fixity to skin or deep tissues, skin ulceration or trismus.**
- **Cervical lymphadenopathy.**
- **A lump in the submandibular, sublingual or minor salivary glands,** where the proportion of malignant tumours is higher.
- **A parapharyngeal mass** pushing the tonsil medially (deep lobe tumour).
- **Sjögren's syndrome with a new persistent swelling:** lymphoma.

CHAPTER TWO

2 History and examination

Which gland, and why?

2.1 History

Ask how long the swelling has been present, whether it fluctuates or is constant, whether it is related to eating, whether it is painful, and whether it is one gland or several. Ask about a bad taste or pus in the mouth, dry eyes and dry mouth, joint pains, fever, weight loss and night sweats, alcohol intake, diabetes, medications (anticholinergics, diuretics), radiotherapy, HIV risk, and previous skin cancers of the face and scalp, which metastasise to parotid nodes.

2.2 Examination

Inspect and palpate the gland, bimanually for the submandibular gland and the floor of mouth, feeling for a stone along Wharton's duct. Look at the duct openings: the parotid (Stensen's) duct opposite the upper second molar and the submandibular duct beside the frenulum; massage the gland to see whether clear saliva, cloudy saliva or pus comes out. Test every branch of the facial nerve, examine the oropharynx for medial displacement of the tonsil, check the neck for nodes, and inspect the skin of the face and scalp.

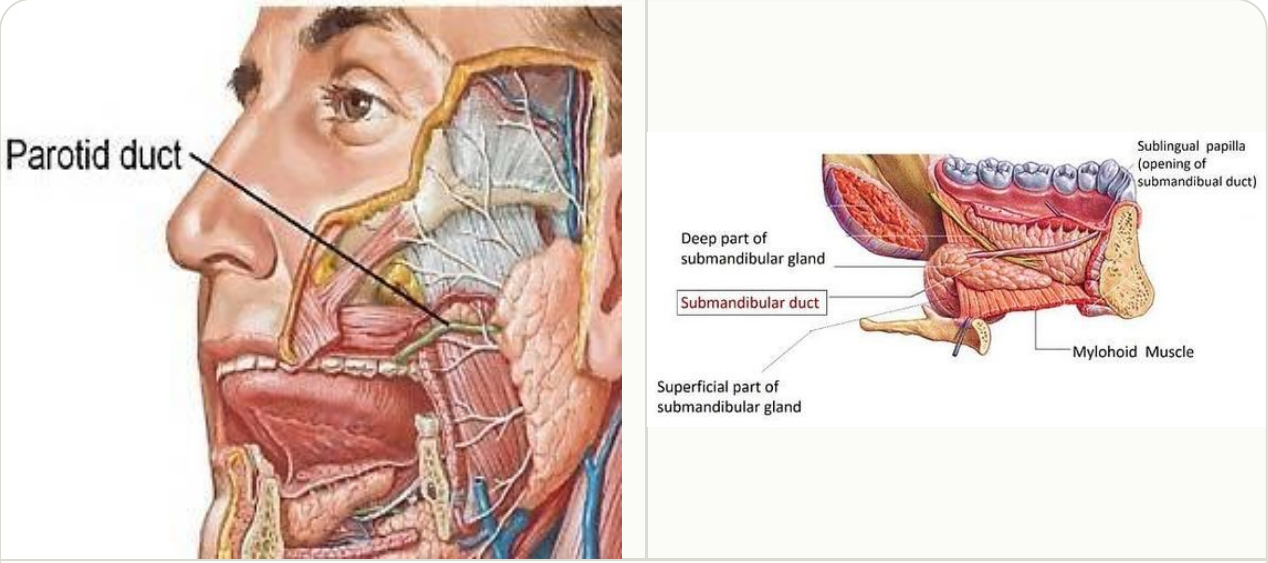


FIGURE 1 First, the parotid gland and Stensen's duct. Second, the sublingual and submandibular glands with their ducts.

Anatomy atlas · ENT Steps p. 181, ENT Steps p. 191

3 CHAPTER THREE Investigation

Ultrasound first.

3.1 Imaging and tissue

Ultrasound is the first-line investigation for almost all salivary swellings. It shows stones, duct dilatation, sialectasis, diffuse inflammatory change, cysts and solid masses, and guides **fine-needle aspiration cytology** or **core biopsy**. Cytology is reported using the Milan system, which grades the risk of malignancy; core biopsy

gives more tissue when cytology is inconclusive or lymphoma is suspected. **MRI** defines the extent of a tumour, deep lobe involvement and perineural spread; **CT** shows stones and bony invasion; and **sialography** or **MR sialography** shows ducts and strictures. **Sialendoscopy** is both diagnostic and therapeutic for obstruction. Blood tests include inflammatory markers, autoantibodies (anti-Ro, anti-La) for Sjögren's syndrome, serum IgG4, ACE and HIV when indicated.

4 CHAPTER FOUR Causes

Obstruction, infection, systemic, neoplasm.

4.1 Stones and strictures

About 80 to 90% of salivary stones occur in the submandibular gland, because its saliva is more alkaline and mucous and its duct runs uphill. Patients have **meal-time syndrome**: pain and swelling with eating that settle over hours. Stones near the duct opening are removed transorally; stones up to about 4 to 5 mm can be retrieved with a sialendoscope basket, and larger ones fragmented by laser or extracorporeal lithotripsy or removed by a combined endoscopic and open approach. Gland excision is reserved for failure. Parotid problems more often come from strictures, which are dilated endoscopically.

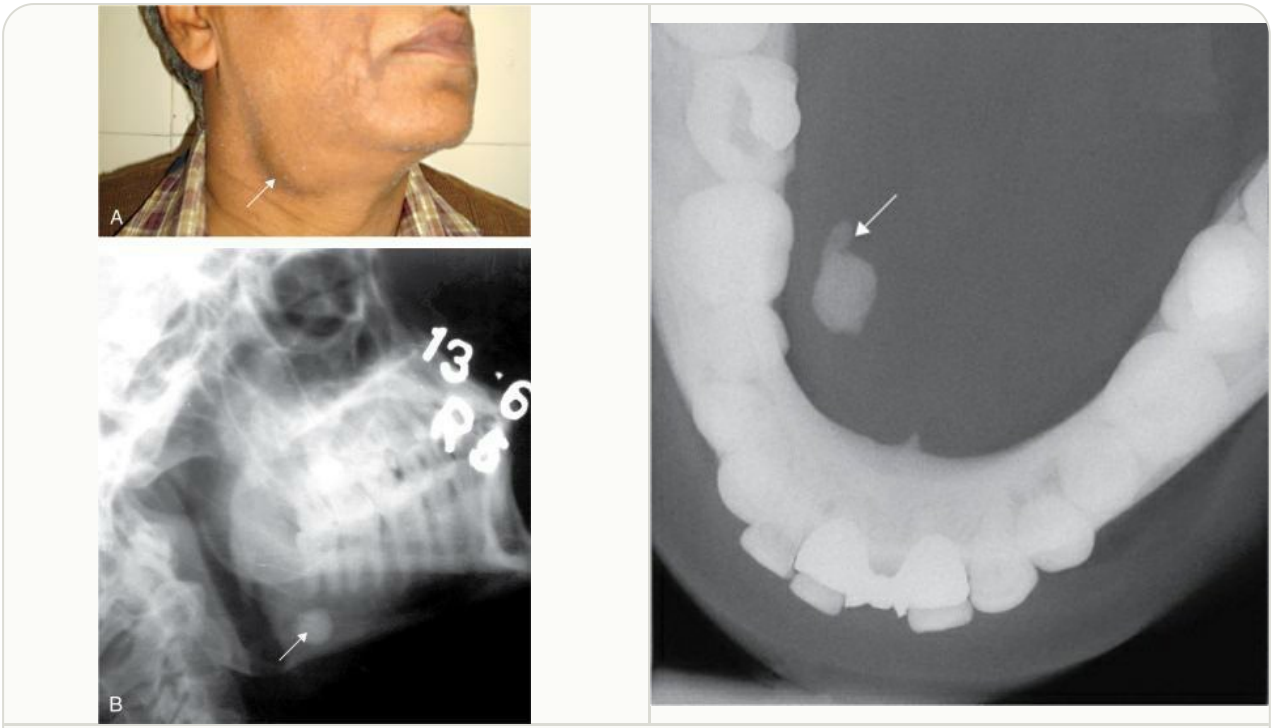


FIGURE 2 Submandibular sialadenitis, and a radio-opaque stone in the submandibular duct.
Dhingra, figs 45.2 and 45.3

4.2 Infection

Acute bacterial sialadenitis, usually *Staphylococcus aureus*, occurs in dehydrated, elderly or postoperative patients with reduced salivary flow; treatment is rehydration, sialogogues, massage, oral hygiene and antibiotics, with drainage of any abscess. **Mumps** causes bilateral painful parotitis with malaise and is managed supportively. **Juvenile recurrent parotitis** causes repeated parotid swellings in children and usually resolves by puberty.

4.3 Systemic and infiltrative causes

Sjögren's syndrome causes dry eyes, dry mouth and parotid swelling, and carries a markedly raised risk of MALT lymphoma. **IgG4-related disease** causes firm bilateral submandibular swellings (Küttner tumour) that respond to steroids. **Sarcoidosis** can cause parotid swelling; with uveitis, fever and facial palsy it forms Heerfordt's syndrome. **HIV** causes lymphoepithelial cysts. **Sialosis** is a painless bilateral enlargement associated with alcohol, diabetes, malnutrition and drugs.

4.4 Tumours

Most salivary tumours arise in the parotid, and about 80% of parotid tumours are benign, most commonly **pleomorphic adenoma**, followed by **Warthin's tumour** (often bilateral, in older male smokers). The smaller the gland, the more likely a tumour is malignant: about half of submandibular tumours and most sublingual and minor salivary gland tumours are malignant. Malignant tumours include mucoepidermoid carcinoma (the commonest), adenoid cystic carcinoma (with perineural spread), acinic cell carcinoma, salivary duct carcinoma and carcinoma ex pleomorphic adenoma, as well as metastases from cutaneous squamous cell carcinoma and melanoma in parotid nodes. Benign parotid tumours are removed by superficial or partial parotidectomy, or extracapsular dissection, preserving the facial nerve; malignant tumours are managed by the head and neck multidisciplinary team.

THE SWOLLEN SALIVARY GLAND

1 **Pattern**

- → **meal-related**
  stone or stricture — ultrasound, sialendoscopy
- → **acute painful with pus**
  bacterial sialadenitis — fluids, antibiotics, drain abscess
- → **bilateral painless**
  bloods for Sjögren's, IgG4, sarcoid, HIV
- → **discrete lump**
  step 2

2 **Discrete lump**

- → ultrasound with fine-needle cytology or core biopsy
- → MRI if malignant, deep or fixed

3 **Red flags?**

- → **facial weakness, pain, fixity, nodes**
  treat as malignant — MDT
- → **benign cytology**

parotidectomy or extracapsular dissection, or observation in selected patients

4 **Sjögren's with persistent swelling**

→ biopsy to exclude lymphoma



FIGURE 3 Squamous cell carcinoma of the parotid: (A) a parotid swelling and (B) facial palsy, the red flag.
Dhingra, fig. 46.1

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Pattern recognition:** meal-time, acute, bilateral and discrete swellings.
- **Why stones form in the submandibular gland.**
- **Ultrasound and fine-needle cytology** as first-line tests; the Milan system.
- **The proportion of malignancy** by gland.
- **Facial palsy with a parotid lump** means malignancy.

- **Sjögren's syndrome and lymphoma risk.**

5.2 Common pitfalls

COMMON PITFALLS

- Performing an open biopsy of a parotid lump, which risks seeding a pleomorphic adenoma and the facial nerve.
- Missing a cutaneous primary in a patient with a parotid node.
- Removing a submandibular gland for a stone that could be removed endoscopically.

Written for this app from Oxford Textbook of Otolaryngology ch. 58, 82, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Oxford Specialist Handbook ch. 7, Dhingra ch. 45, 46. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialolith of the Submandibular Gland

IMAGING FINDINGS

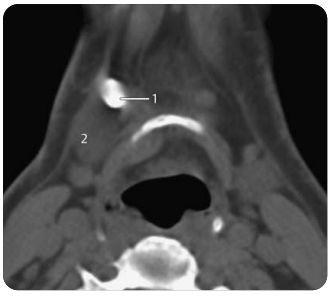
- Dense calculus in the Wharton duct or hilum on non-contrast CT
- Dilated duct proximal to the stone
- Enlarged inflamed or atrophic gland

TOP DIFFERENTIALS

- Phlebolith (venous malformation)
- Calcified node
- Dental fragment

CHECKLIST

- Position relative to the mylohyoid and hilum
- Ultrasound shows stone and duct without radiation



CT, axial bone window.A hyperdensity (1), with similar density as bone, suggestive of a sialolith (stone, calculus), probably obstructin outflo…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialadenitis

IMAGING FINDINGS

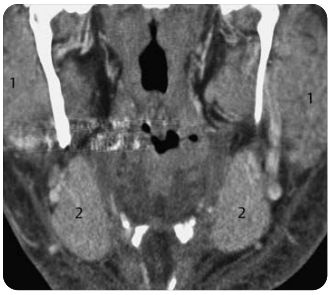
- Enlarged, enhancing gland with stranding of the surrounding fat
- Dilated duct, often with a stone
- Abscess: rim-enhancing collection

TOP DIFFERENTIALS

- Dental infection
- Nodal metastasis
- Gland tumour

CHECKLIST

- Look for a stone along the duct
- Chronic: small, fibrotic gland



Patient with chronic intermittent enlargement of the salivary glands and diffuse pain.
Dunnebier, Imaging for…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · TUMOURS

Pleomorphic Adenoma of the Parotid Gland

IMAGING FINDINGS

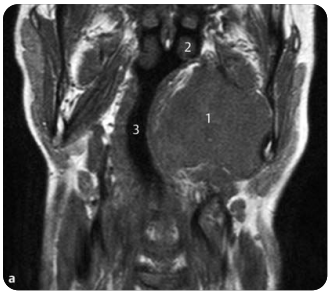
- Well-defined, lobulated mass with a capsule
- Very bright on T2
- Deep lobe: widens the stylomandibular gap, pushes parapharyngeal fat medially

TOP DIFFERENTIALS

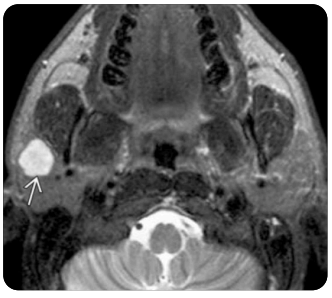
- Warthin tumour (tail, multiple, smokers)
- Low-grade mucoepidermoid carcinoma
- Intraparotid node

CHECKLIST

- Superficial or deep to the retromandibular vein
- Never enucleate: recurrence



MRI, T1-weighted, coronal. Note the extensive, isointense, smooth-bordered lesion (1) in the hypopharyngeal area: from the level o…



Axial T2 FS MR reveals a sharply circumscribed, high-signal BMT in the superficial lobe of the parotid gland. This tumor at the very least abuts, if…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88.

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

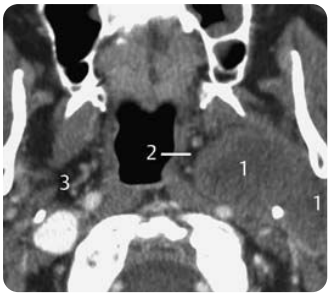
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

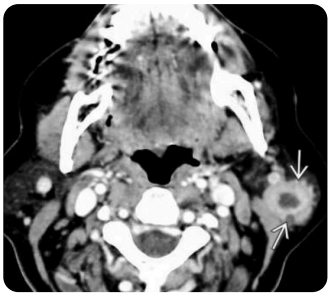
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.